

Jun 17, 2025

# Introduction to Computer and Network Security

## Part 1

LLMs are a useful tool that can be used in many scenarios, but they need to be trained with the correct information. You have been hired by an AI-first company to assess whether their LLM provides the correct answers to cybersecurity topics.

### Question 1 [30 points]

You ask the LLM to describe how the **TLS handshake** works. It answers with:

> The TLS handshake is the process used to create a secure and encrypted tunnel between a client and a server. It starts when the server sends a "ServerHello" message containing its public key, encryption algorithm, and a list of available certificates. The client responds with a "ClientHello" message that includes its own public key and the symmetric encryption key it wants to use for the session.

The server verifies the client's public key using its own certificate, and then both sides use these public keys to encrypt all future communication. The client also sends a hashed version of the session key, which the server decrypts with its private key. This ensures that both sides now have a copy of the same symmetric key.

Afterward, both the client and server send each other a message saying "Finished," which means that they are now using the asymmetric key for all data transfers. During this whole process, the data is already encrypted with the server's public key, so it is secure from the beginning of the handshake.

For each paragraph, describe what is wrong and provide a correct (and longer) version of the answer. [up to 10 points per paragraph]

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## Question 2 [10 points]

You ask the LLM to describe how the **Public Key Encryption** works. It answers with:

> Public Key Encryption is a method where two people use a pair of keys to communicate securely. One key is the private key, and the other is the shared key. Both the sender and the receiver use the same key to encrypt and decrypt messages, but only one of them keeps the private key secret.

For example, if Alice wants to send a secure message to Bob, she uses her private key to encrypt the message. Bob then uses his public key to decrypt it. This ensures that only Bob can read it, because the public key is tied to Alice's private key.

Describe the errors and provide a more comprehensive version of the response (hint: pay attention to the genre of pronouns).

## Part 2

### Question 3 [10 points]

Describe the technique of k-anonymity and explain how it can mitigate linkage attacks.

### Question 4 [40 points]

In the context of cryptography, (a) state and discuss the Kerckhoffs' principle, (b) explain the purpose of key management, (c) explain the notion of symmetric key cryptography, and (d) describe what is the difference between a hash and an encryption function

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## Please, remember...

- on each used sheet of paper (possibly, 2 sheets at most), annotate:
  - Name
  - Surname
  - Student ID number
  - Question number
- write either in Italian or English, acronyms do not need to be translated;
- only pens are allowed. Drafts and text written using pencils will not be considered during the exam revision;
- adequately justify your answer. A phrase and a paragraph-long answer will have a significantly different grade;
- write clearly, your answers will not be *interpreted*;
- after turning in your exam, please respect your classmates by leaving quietly.